

Cat Hauser

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EDUCATION

Boston University

Boston, MA

Ph.D. Student

September 2024 – Present

- Working in the Kaufman-Deyle Systems Ecology Lab
- Area of interest: combining a quantitative, modeling, and field work approach to analyze coral and coral reef structure, systems, and dynamics
- GPA: 3.97 out of 4.0

BU URBAN Trainee

- Member of the Graduate Program in Urban Biogeoscience and Environmental Health (BU URBAN).
- Program includes interdisciplinary training and experience with integrating research into policy and governance.

B.S. in Biomedical Engineering

Rochester, NY

University of Rochester

2016-2020

Concentration in Biosystems and Signals

Minor in Studio Art

- GPA: 3.84 out of 4.00

PUBLICATIONS AND GRANTS

Publications

Rahman, S.M., **Hauser, C.**, Faucher, S., Fine, E., Luebke, A.E. 2024. A Vestibular Challenge Combined with Calcitonin Gene-Related Peptide (CGRP) Promotes Anxiety-Like Behaviors. *eNeuro* 11(7): ENEURO.0270-23.2024. doi: 10.1523/ENEURO.0270-23.2024.

Rahman, S.M., **Hauser, C.**, Luebke A.E. 2024. Loss of calcitonin gene-related peptide (α CGRP) and use of a vestibular challenge highlight balance deficiencies in aging mice. *PLoS One* 19(6): e0303801. doi: 10.1371/journal.pone.0303801.

Posters

Cat Hauser, Lisa Carne, Art Gleason, Les Kaufman, Eva Romero, Ethan Deyle. The Pixels of Protection: Revealing Hidden Geometry of Corals Using Deep Learning and Photogrammetry to Support Reef Restoration. [Poster Presentation]. 16th International Coral Reef Symposium (ICRS 2026), Auckland, New Zealand.

Magaly Koch, [...] **Catherine Hauser**, [...] Gayatri Marliyani; *OS51E-0456 Global Coastal Hazard Perspectives: Insights from the 2025 Advanced Studies Institute Graduate Training Program on Multidisciplinary Coastal Zone Hazards in Java, Indonesia*
<https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1925675>

Grants and Scholarships

2026

Conference Grant – sponsored by Boston University Graduate Student Organization \$1000 USD

2025

Travel Award – sponsored by Boston University Biology Department \$500 USD

Challenge Grant – Naturally Mimicking Hybrid Reefs, sponsored by NSF Strong NSF \$15,000 USD

Advanced Dive Training Grant - Professional Diving, sponsored by Women Divers Hall of Fame \$1000 USD

Graduate Research Abroad Fellowship (GRAF) – Summer Research, sponsored by Boston University \$4000 USD

RESEARCH EXPERIENCE

BU URBAN Internship

Caye Caulker, Belize

Boston University/Fragments of Hope/University of South Florida

August 2025 – August 2026

- Organized and led a workshop in Caye Caulker, Belize on February 24-25, 2026, with the help of mentors Ali Cansino (Fragments of Hope) and Alex Webb (University of South Florida) and the [NSF Strong Coasts project](#). This workshop engaged the community on the potential use of hybrid reefs in the area. This [video](#) shows a recap of the workshop.
- Analyzed feedback results to create an accessible [Story Map](#) detailing the workshop and site visits.
- Created a comprehensive public facing project report. This report includes a workshop overview, the data analysis, and preliminary monitoring and safety plans.
- Shared results with the Belize Fisheries Department and distributed to the Caye Caulker community.
- Presented a summary of the work to the Belize Fisheries Department at the annual NSF Strong Coasts project meeting in June 2026.

Ph.D. Field Work

Boston University/Fragments of Hope

Placencia, Belize

July and December 2025

- Organized and led a field expedition in Placencia, Belize to collect research data for my Ph.D. work and for Fragments of Hope. This included setting up the accommodations, organizing the boat schedule and managing a grant budget.
- Created a dive plan and used SCUBA and free diving to collect data.
- Took macro photos of *Acropora palmata* and organized the photos to be analyzed for research.
- Took 3D videos of *Acropora palmata* to be analyzed for research and modeling.

Advanced Studies Institute: Coastal Zone Hazards Fellowship

Boston University

Java, Indonesia

June 2025

- Participated in a two-week field work program in Java, Indonesia.
- First-hand experience studying coastal hazards including flooding, volcanic activity, erosion and land subsidence through site visits and engagement with the community.
- Learned how techniques such as remote sensing, SAR, GIS, and Deep Learning can be used to study coastal hazards.
- Created a research proposal presentation to conclude the fellowship collaborating with other fellowship members.

Coral Reef Fishes Course Field Work

Boston University

Calabash Caye Field Station, Belize

December 2024

Professor: Dr. Peter Buston

- Led a team of five using SCUBA and snorkeling methods to collect 30-minute timed transects recording *Chaetodontidae* family pairing observations.
- Analyzed and plotted the data in R.
- Created a report and presentation on the findings.

Coastal Biogeochemistry Course Field Work

Boston University

Waquoit Bay, Cape Cod

October 2024

Professor: Dr. Robinson Fulweiler

- Worked in a team of three to measure carbon dioxide and methane fluxes using a Picarro GasScouterTM G4301 Mobile Gas Concentration Analyzer.
- Investigated three species of salt marsh vegetation: *Spartina alterniflora*, *Juncus gerardii*, and *Distichlis spicata*.
- Created scripts and analyzed the data in both R and Python.

Research Assistant, Dr. Anne Luebke's Lab

University of Rochester's Medical Center

Rochester, NY

January 2017 – May 2020

- Led projects investigating vestibular motion and migraines studying the effect of provocative motion in calcitonin gene-related peptide receptor animals using postural sway inducing apparatuses and use of MATLAB for data analysis.
- Lab techniques included Rotarod Vestibular Challenge procedures, Postural Sway procedures, animal husbandry, PCR and DNA gel electrophoresis to observe mouse genetics, behavioral postural and provocative testing, and animal injections certified for work with mice.

Research Assistant, Isam Khalaila's Research Group

Ben Gurion University

Be'er Sheva, Israel

June 2018 – August 2018

- Investigated cellular/molecular mechanisms for human breast cancer and prostate cancer cells communication to see what factors affect the process of O-GlcNAcylation in cancer cells.
- Led cell culture for human prostate and breast cancer cells in the lab and performed standard lab protocols including Western Blots, LTX DNA transfection, and PCR analysis.

INDUSTRY EXPERIENCE

Uplight

Software Engineer I

Implementation Engineer

Denver, CO

May 2023 - August 2024

March 2022 - May 2023

- Worked as a backend Python Software Engineer.
- Responsible for assisting with the creation of APIs to surface data at Uplight internally and to external customers.
- Responsible for completing efficient Python scripts and writing unit and functional tests.
- Responsible for expanding my knowledge of infrastructure and engineering best practices.
- Worked to implement the Uplight Software for utility companies to analyze and optimize energy usage.
- Responsible for implementing a widget solution for non-residential energy users.
- Responsible for Implementation Engineer automation and tooling creation, including updating databases transferring Uplight commonly used API calls into Postman and Python scripts, and providing documentation for these processes.

Ambra Health

Professional Services Senior Analyst

Customer Success Analyst

Denver, CO

April 2021 - March 2022

July 2020 - April 2021

- Led highly technical projects for Ambra customers to provide smooth medical image exchange between facilities.
- Matched workflows to customer needs, as well as set up specialized configurations.
- Project highlights include: Worked with UNOS research project to develop a Python script that calls to the Ambra API, worked with Mayo Clinic research to utilize the Ambra AI stack for anonymization, worked with research companies to integrate the Ambra API into their software, worked with Children's Hospital of Philadelphia to connect Ambra to a GCP bucket, set up patient portal for hospitals, set up single sign-on for hospitals, and assisted with the Ohio State University implementation in both a public and private cloud.

TEACHING AND MENTORSHIP EXPERIENCE

Tropical Marine Seagrass Teaching Fellow

Boston University

Assisting Dr. Alyssa Novak

Boston, MA

November 2025– December 2025

- Responsible for assisting students with field work research projects at Calabash Caye Field Station in Belize. Including assisting with field work research project design and execution in the field.
- Responsible for assisting students with species ID including Caribbean corals, fishes, seagrass, sponges, invertebrates, and megafauna.
- Responsible for assisting students with R studio code and statistical analysis.

Quantitative FishRies Teaching Fellow

Boston University

Assisting Dr. Ethan Deyle

Boston, MA

October 2025– November 2025

- Responsible for assisting students with R code for fisheries modeling labs.
- Models Labs included: Generalized Additive Modeling (GAM), Empirical Dynamic Modeling (EDM), Population Dynamics Modeling, Leslie Matrix Modeling.

RISE Program Mentor

Boston University

Boston, MA

July 2025- August 2025

- Assisted in mentoring an upcoming high school senior with a summer research project.
- Supported the mentee in coding work for the project and with creating a final poster.

VOLUNTEER WORK

Denver Aquarium

Pathway Guide

Denver, CO

July 2022 – August 2024

- Recipient of the *Volunteer Appreciation Award* in April 2024
- Educated guests about all animals and exhibits at the aquarium.
- Answered any questions about the aquarium.
- Attended learning sessions and educational talks.
- Assisted feeding the stingrays at the touch pool and informing guests of the proper feeding technique.

SKILLS AND CERTIFICATION

- ***Languages and Software*** – R, Python, MATLAB, JSON, Postman, RESTful APIs, MySQL, PostgreSQL, Java, Splunk, JIRA, HTML, regex, Excel, AWS Cloud Practitioner Course, GCP experience, BigQuery, GitHub, Agisoft
- ***Skills*** – scripting, data reconciliation, control systems design, data analysis, 3D printing, circuit design, linear algebra, differential equations, multidimensional calculus
- ***Diving Certifications*** – AAUS Scientific Diver and Lead Diver – 100 fsw Depth Authorization